# 口腔诊所管理系统｜逻辑 ERD ·业务局部图

- 诊所、岗位与基础资料
- 患者、预约与诊疗
- 收费明细、支付与催收
- 义齿委托、交付与通知

### encounter｜诊疗记录（引用）

| 键 | 字段 | SQL 类型 | 外键目标 |
|---|---|---|---|
| PK | id | BIGINT | |

### employee｜员工（引用）

| 键 | 字段 | SQL 类型 | 外键目标 |
|---|---|---|---|
| PK | id | BIGINT | |

### basic_option｜分类基础选项（引用）

| 键 | 字段 | SQL 类型 | 外键目标 |
|---|---|---|---|
| PK | id | BIGINT | |

### product｜义齿产品

| 键 | 字段 | SQL 类型 | 外键目标 |
|---|---|---|---|
| PK | id | BIGINT | |
| UK | product_code | VARCHAR(32) | |
| | name | VARCHAR(80) | |
| | material | VARCHAR(80) | |
| FK | unit_id | BIGINT | basic_option.id |
| | specification | VARCHAR(240)? | |
| | requires_attachment | BOOLEAN | |
| | active | BOOLEAN | |

### dental_order｜义齿委托订单

| 键 | 字段 | SQL 类型 | 外键目标 |
|---|---|---|---|
| PK | id | BIGINT | |
| UK | order_no | VARCHAR(32) | |
| FK | encounter_id | BIGINT | encounter.id |
| FK | product_id | BIGINT | product.id |
| FK | lab_id | BIGINT | dental_lab.id |
| FK | created_by | BIGINT | employee.id |
| FK | color_id | BIGINT | basic_option.id |
| FK | box_color_id | BIGINT? | basic_option.id |
| FK | type_id | BIGINT | basic_option.id |
| FK | unit_id | BIGINT | basic_option.id |
| | quantity | DECIMAL(10,2) | |
| | repair_requirement | TEXT | |
| | promised_date | DATE | |
| | stage | VARCHAR(24) | |
| | dispatch_result | VARCHAR(24) | |
| | expected_at | TIMESTAMP? | |
| | created_at | TIMESTAMP | |
| | sent_at | TIMESTAMP? | |
| | accepted_at | TIMESTAMP? | |
| | trial_at | TIMESTAMP? | |
| | trial_result | VARCHAR(24)? | |
| | delivered_at | TIMESTAMP? | |
| | cancelled_at | TIMESTAMP? | |
| | cancel_reason | TEXT? | |

### dental_lab｜合作技工所

| 键 | 字段 | SQL 类型 | 外键目标 |
|---|---|---|---|
| PK | id | BIGINT | |
| UK | lab_code | VARCHAR(32) | |
| | name | VARCHAR(120) | |
| | contact | VARCHAR(80) | |
| | phone | VARCHAR(32) | |
| | address | VARCHAR(240)? | |
| | interface_code | VARCHAR(64)? | |
| | active | BOOLEAN | |

### order_tooth｜订单牙位

| 键 | 字段 | SQL 类型 | 外键目标 |
|---|---|---|---|
| PK,FK | order_id | BIGINT | dental_order.id |
| PK | tooth_code | VARCHAR(8) | |
| | repair_note | VARCHAR(240)? | |

### order_event｜技工所回执与进展

| 键 | 字段 | SQL 类型 | 外键目标 |
|---|---|---|---|
| PK | id | BIGINT | |
| FK,UK* | order_id | BIGINT | dental_order.id |
| UK* | source | VARCHAR(64) | |
| UK* | event_no | VARCHAR(64) | |
| FK | changed_by | BIGINT? | employee.id |
| | old_stage | VARCHAR(24) | |
| | new_stage | VARCHAR(24) | |
| | reason | TEXT? | |
| | external_order_no | VARCHAR(64)? | |
| | occurred_at | TIMESTAMP | |
| | received_at | TIMESTAMP | |

### order_attachment｜修复附件

| 键 | 字段 | SQL 类型 | 外键目标 |
|---|---|---|---|
| PK | id | BIGINT | |
| FK | order_id | BIGINT | dental_order.id |
| FK | type_id | BIGINT | basic_option.id |
| FK | uploaded_by | BIGINT | employee.id |
| | file_name | VARCHAR(180) | |
| UK | storage_key | VARCHAR(240) | |
| | mime_type | VARCHAR(80) | |
| | byte_size | BIGINT | |
| | checksum | VARCHAR(64)? | |
| | uploaded_at | TIMESTAMP | |

### reminder｜预约与订单通知

| 键 | 字段 | SQL 类型 | 外键目标 |
|---|---|---|---|
| PK | id | BIGINT | |
| FK,UK* | order_id | BIGINT? | dental_order.id |
| FK,UK* | appointment_id | BIGINT? | appointment.id |
| FK | channel_id | BIGINT | basic_option.id |
| FK | created_by | BIGINT | employee.id |
| UK* | kind | VARCHAR(24) | |
| UK* | scheduled_at | TIMESTAMP | |
| UK* | recipient | VARCHAR(120) | |
| | content | TEXT | |
| | delivery_result | VARCHAR(24) | |
| | attempt_count | INTEGER | |
| | last_attempt_at | TIMESTAMP? | |
| | sent_at | TIMESTAMP? | |
| | failure_reason | TEXT? | |

PK：主键；FK：外键；UK：唯一；UK*：联合唯一；?：可空。实线三角箭头：外键表 → 被引用表；同一对表的多个外键合并一条线，逐字段目标见表内。